

TOPIC NAME: Fall 2024 (A)

DAY: _____

TIME: _____

DATE: / /

① a)

Broadcast address: 42.1.63.255

Maximum hosts supported: 16382

$$2^h - 2 = 16382$$

$$\Rightarrow 2^h = 16384$$

$$\Rightarrow h = 14$$

$$\text{Prefix length} = 32 - 14 = 18$$

1 1111111 1111111 11000000 00000000

255 255 192 0

(A) 1510.0.0.0

Subnet mask: 255.255.192.0

(B) 2510.8.1.64

(C) 2510.8.1.64

(D) host = 14

(E) 00101010 00000001 00111111 11111111

network address →

00101010 00000001 00000000 00000000

42

1

0

0

Network address: 42.1.0.0 / 18

TOPIC NAME :

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(iii) Network address : 42.1.0.0/18

Name	Host	Host + 2	Block Size	host bits	prefix
A	2000	2002	2046	11	21
B	1022	1024	1024	10	22
C	512	514	1024	10	22
D	2	4	4	2	30
E	2	4	4	2	30

net

00000000	00000000	11111111	11111111
00000000	00000000	00000000	00000000
00000000	00000000	00000000	00000000
00001000	00000000	00000000	00000000
00001000	00000000	00000000	00000000
00001100	00000000	00000000	00000000
00010000	00000000	00000000	00000000
00010000	00000000	00000000	00000000
00010000	00000000	00000000	00000000

00000000 → 0.0/18
00000000 → 0.0/21 (A)
00000000 → 42.1.8.0/21
00000000 → 42.1.8.0/22 (B)
00000000 → 42.1.12.0/22 (C)
00000000 → 42.1.16.0/21
00000000 → 42.1.16.0/30 (D)
00000100 → 42.1.16.4/30 (E)

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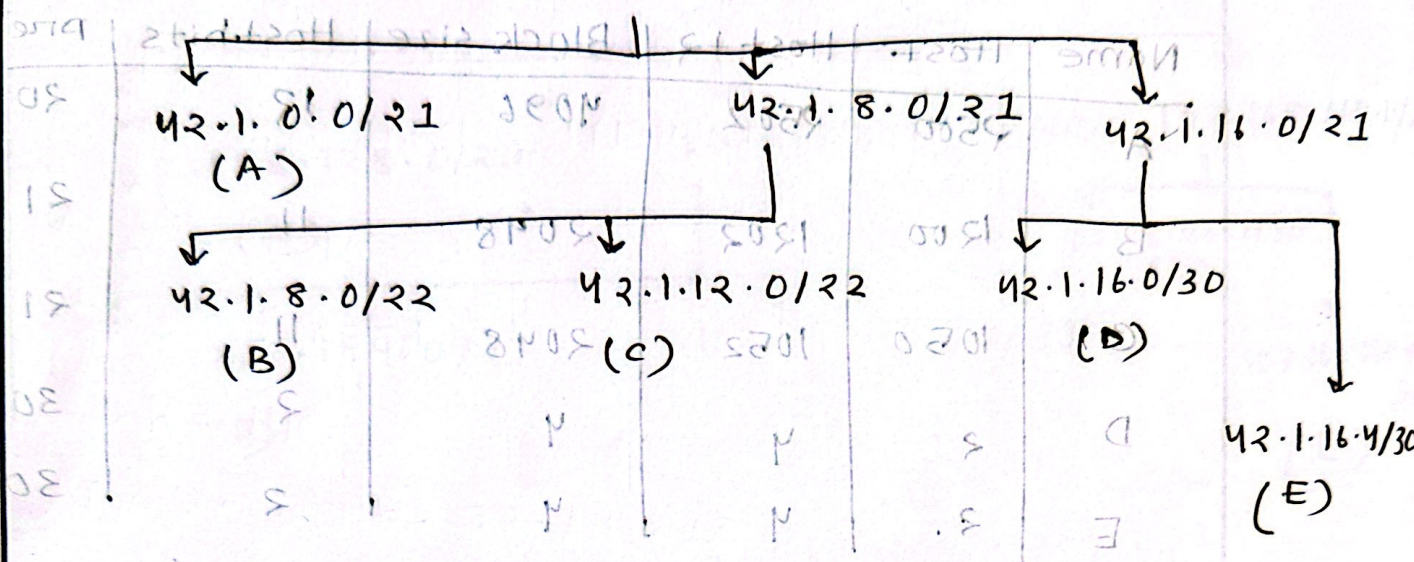
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42.1.0.0/18



Summer 2024 (A)

① a) Prefix mask = 17

host = 32 - 17 = 15

11111111 11111111 10000000

255

255

128

Subnet mask: 255.255.128.0

b) IPv4 address: 107.168.177.108/17

01101011 10101000 10110001 01101100

Network address =

01101011 10101000 10000000 00000000

107

168

128

0

Network address: 107.168.128.0

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c) Network address: 10.7.168.128/17

Name	Host	Host + 2	Block size	Host bits	Prefix
A	2500	2502	4096	13	20
B	1200	1202	2048	11	21
C	1050	1052	2048	11	21
D	2	4	4	2	30
E	2	4	4	2	30

1000|0000 00000000 → 128.0/17
 1000|0000 00000000 → 128.0/20 (A)
 1001|0000 00000000 → 144.0/20
 10010|000 00000000 → 144.0/21 (B)
 10011000 00000000 → 152.0/21 (C)
 10100000 00000000 → 160.0/20
 10100000 00000000 → 160.0/30 (D)
 10100000 00000100 → 160.4/30 (E)

FL/201-FFI.201.FOI: 201101 101

00110110 1001101 00010101 11010110

Network address

00000000 00000000 00010101 11010110

20

251

201

FOI

0.251.201.FOI: 201101 101

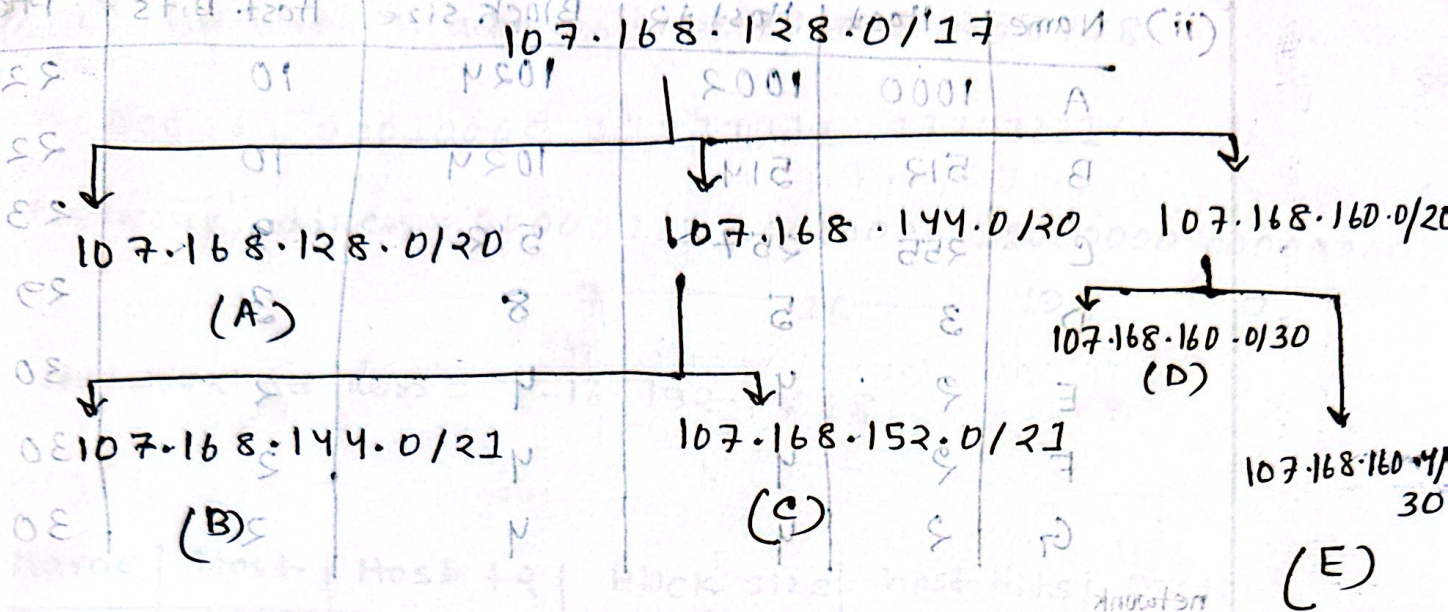
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SPRING 24 (A)

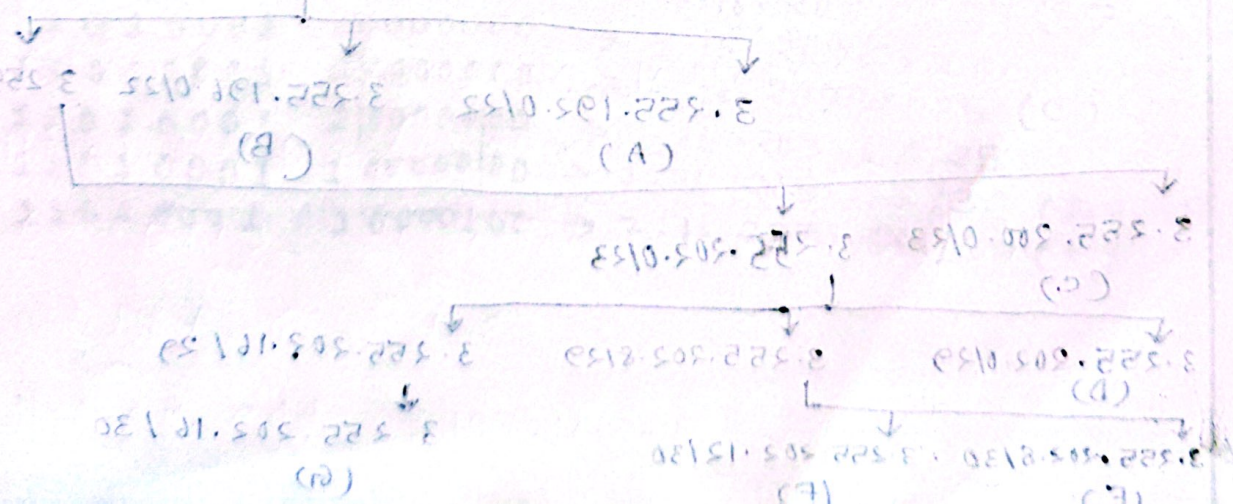
① i)

Host bits = $32 - 19 = 13$ bits

Total IP = $2^{13} = 8192$ address

If we assume the smallest subnet is a /30 (used for router to router links), the number of bits available for subnetting is $30 - 19 = 11$ bits

maximum number of subnets = $2^{11} = 2048$ subnets



TOPIC NAME :

DAY :

TIME :

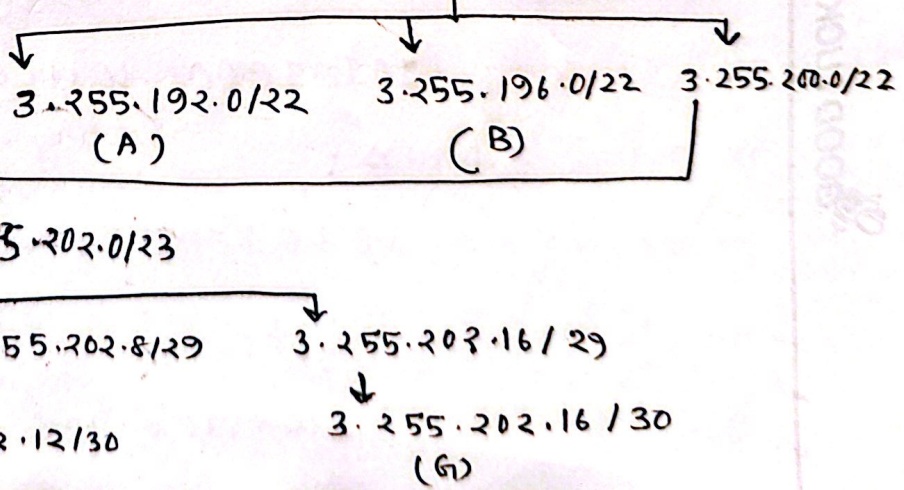
DATE :

(ii)	Name	Host	Host +2	Block size	Host Bits	Prefix
	A	1000	1002	1024	10	22
	B	512	514	1024	10	22
	C	255	257	512	9	23
	D	3	5	8	3	29
	E	2	4	4	2	30
	F	2	4	4	2	30
	G	2	4	4	2	30

network

11000000 00000000 → 192.0/19
 11000000 00000000 → 192.0/22 (A)
 11000100 00000000 → 196.0/22 (B)
 11001000 00000000 → 200.0/22 (C)
 11001000 00000000 → 200.0/23 (C)
 11001010 00000000 → 202.0/23 (D)
 11001010 00001000 → 202.8/29 (D)
 11001010 00001000 → 202.8/30 (E)
 11001010 00001100 → 202.12/30 (F)
 11001010 00010000 → 202.16/30 (G)

3.255.192.0/19



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TOPIC NAME: Spring 2025 (A)

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① i) Broadcast address = 7.16.192.255/18

00000111 00010000 11111111 11111111

Network address = 00000111 00010000 11000000 00000000

7

16

192

0

Network address = 7.16.192.0/18

(ii)

Name	Host	Host + 2	Block size	host bits	Prefix
A	2501	2503	4096	12	20
B	128	130	256	8	24
C	65	67	128	7	25
D	2	4	4	2	30
E	2	4	4	2	30

net

11000000 00000000 → 7.16.192.0/18

11000000 00000000 → 7.16.192.0/20 (A)

11010000 00000000 → 7.16.208.0/20

11010000 00000000 → 7.16.208.0/24 (B)

11010001 00000000 → 7.16.209.0/24

11010001 00000000 → 7.16.209.0/25 (C)

11010001 10000000 → 7.16.209.128/25

11010001 10000000 → 7.16.209.128/30 (D)

11010001 10000100 → 7.16.209.132/30 (E)

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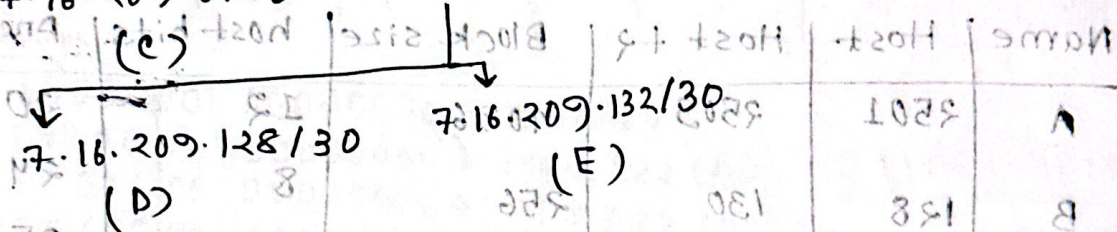
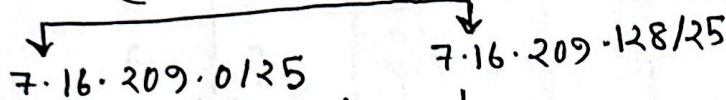
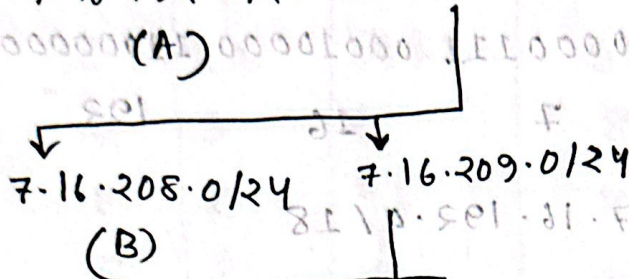
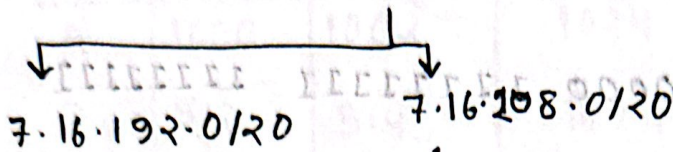
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7.16.192.0/18



(iii) $128 - 67 = 61$
Summen 25 (A)

① i) Sheldon: 192.128.0.2 (2nd usable IP)

Leonard: 192.128.255.253 (2nd last usable IP)

Network address: 192.128.0.0

Broadcast address: 192.128.255.255

The range: 192.128.0.0 to 192.128.255.255

prefix mask: /16

ii) Howard wants to send a packet to all devices on R1-LAN (a directed broadcast).

So, the broadcast address for R1-LAN is 192.128.255.255

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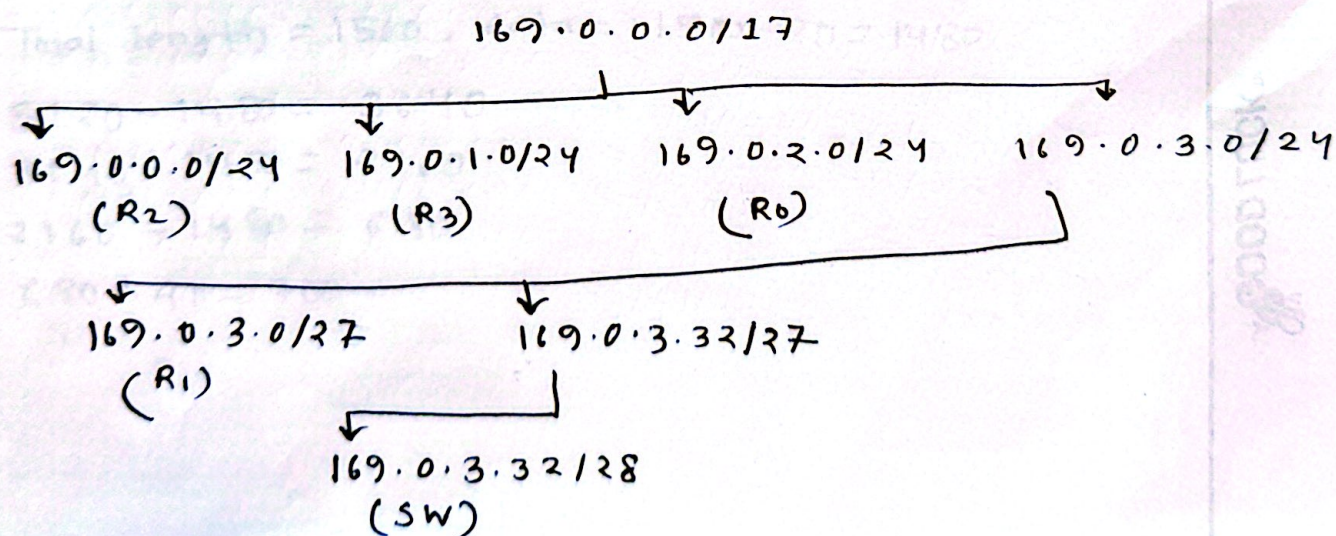
TIME: _____

DATE: / /

(ii) 169.0.0.0/17

Name	Host	Host + 2	Block size	Host bits	Prefix
R ₂	251	253	256	8	24
R ₃	151	153	256	8	24
R ₀	129	131	256	8	24
R ₁	26	28	32	5	27
SW	11	13	16	4	28

$\begin{array}{l} 00000000 \\ 00000000 \\ 00000001 \\ 00000010 \\ 00000011 \\ 00000011 \\ 00000011 \\ 00000011 \end{array} \left| \begin{array}{l} 00000000 \\ 00000000 \rightarrow 0.0/24 \rightarrow R_2 \\ 00000000 \rightarrow 1.0/24 \rightarrow R_3 \\ 00000000 \rightarrow 2.0/24 \rightarrow R_0 \\ 00000000 \rightarrow 3.0/24 \\ 00000000 \rightarrow 3.0/27 \rightarrow R_1 \\ 00100000 \rightarrow 3.32/27 \\ 00100000 \rightarrow 3.32/28 \end{array} \right.$



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(iv) Host + 2 = 11 + 2 = 13

Waste = 16 - 13 = 3

15	3	125	525	135	25
5	8	125	531	121	25
5	8	125	131	133	25
5	7	58	35	25	25
5	4	10	51	11	25

0000000000 0000000000
 1000000000 0000000000
 0000000000 1000000000
 0000000000 0000000000
 0000000000 1000000000
 0000000000 1000000000
 0000000000 1000000000
 0000000000 1000000000
 0000000000 1000000000
 0000000000 1000000000

1000000000

